

Project Architecture, Technical Documentation & Step-by-Step Build Tutorial

A complete technical guide explaining system design, component architecture, modern technology choices, and step-by-step implementation.

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1. Project Vision & Architectural Overview

Sk Sahil's Developer Portfolio is an engineer-grade, high-performance web application designed to demonstrate clean code architecture, system performance, and modern frontend practices. Unlike generic website templates that suffer from bloated JavaScript bundles and slow visual editors, this application was engineered from scratch using **React 19, Vite 8, Tailwind CSS v4, Framer Motion, and Linux CLI interaction models**.

Key engineering achievements of this project include:

- **Signature Expanding Horizontal Navigation:** A custom flex-grow interpolation layout that dynamically expands the active section tab.
- **Embedded Interactive Linux Terminal Modal:** A functional command-line interface featuring input parsing, history navigation (Up/Down arrows), and section shortcuts.
- **Single-Source-of-Truth Data Layer:** Complete decoupling of content from UI components using a centralized ``data.js`` structure.
- **Direct Download Integration:** Built-in HTML5 download handlers for instantaneous PDF resume retrieval.
- **Zero-Bloat Performance:** Ultra-lightweight ~110KB compressed bundle size with sub-second page loads.

Technical Aspect	Generic Template Approach	Sk Sahil's Portfolio Architecture
Bundle & Speed	3MB+ script footprint, heavy DOM trees	~110KB compressed JS, instant Vite compilation
UI Components	Tightly-coupled monolithic HTML	Modular React components with clean props
Interactivity	Basic visual triggers & auto-scroll	Interactive Linux modal terminal CLI emulator
State & Content	Hardcoded text throughout JSX markup	Decoupled data store (<code>`data.js`</code>)

Aesthetic & Styling	Bright agency gradients & visual noise	Minimal slate dark mode inspired by developer tools
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2. Technology Stack & Tool Mechanics

Technology	Tool Functionality & Description	Implementation in this Project
React 19	Industry-standard JavaScript library for building component-based user interfaces.	Serves as the UI foundation (`Navbar`, `ExpandingNav`, `Terminal`, `Profile`, `Projects`). React state (`useState`, `useCallback`) manages active tabs.
Vite 8	Next-generation frontend build engine utilizing native ES modules for ultra-fast compilation.	Provides sub-600ms Hot Module Replacement (HMR) during development and builds optimized production bundles.
Tailwind CSS v4	Utility-first styling framework compiled directly into lightweight CSS.	Configured with custom `@theme` tokens in `src/index.css` defining the Midnight Slate palette, monospace fonts, and hover glow effects.
Framer Motion	Production motion framework for fluid React layout transitions.	Powers hardware-accelerated entrance and exit animations for the Linux Terminal modal window (`AnimatePresence`).
Lucide React & Custom SVGs	Scalable vector icon collection matching modern software standards.	Provides UI icons (Mail, FileText, MapPin) along with custom SVG components for GitHub, LinkedIn, LeetCode, and Vercel in `BrandIcons.jsx`.
Git & GitHub	Distributed version control system and remote repository cloud hosting.	Manages version history at `Sahil-9879/sk_sahil_portfolio` with structured commit milestones.
Vercel Edge Platform	Global edge cloud network optimized for static single-page application hosting.	Automates continuous deployment on every git push to `main` at `sk-sahil-portfolio.vercel.app`.

3. Directory Layout & Modular Structure

sk-sahil-portfolio/ public/ resume.pdf # Direct PDF download file Sk_Sahil_Portfolio_Engineering_Guide.pdf # Technical documentation PDF src/ components/ icons/ BrandIcons.jsx # SVG Brand icons (GitHub, LinkedIn, LeetCode, Vercel) layout/ Navbar.jsx # Header bar with navigation & profile actions		
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■ ■ ■■ sections/	
■ ■ ■ ■■ Profile.jsx	# Hero section & introductory summary
■ ■ ■ ■■ About.jsx	# Background narrative & education timeline
■ ■ ■ ■■ Projects.jsx	# Project cards with GitHub, Vercel & Guide link
s	
■ ■ ■ ■■ TechStack.jsx	# Categorized skills matrix
■ ■ ■ ■■ Contact.jsx	# Contact links grid with hover glow effects
■ ■ ■■ ExpandingNav.jsx	# Signature flex-grow horizontal tab navigation
■ ■ ■■ ContentPanel.jsx	# Dynamic view container
■ ■ ■■ Terminal.jsx	# Linux-style interactive CLI modal window
■ ■■ constants/	
■ ■ ■■ data.js	# Centralized data store (Single Source of Truth)
■ ■■ hooks/	
■ ■ ■■ useTerminal.js	# Custom React hook for CLI state management
■ ■■ App.jsx	# Root layout container
■ ■■ index.css	# Tailwind v4 @theme design tokens
■ ■■ main.jsx	# React 19 DOM mounting entry
■■■ index.html	# Page head, fonts & SEO metadata
■■■ vite.config.js	# Vite configuration & plugin pipeline

4. Step-by-Step Implementation Tutorial

Step 1: Project Scaffolding & Setup

Scaffold a new Vite React application and install styling and motion libraries:

```
npx -y create-vite@latest ./ --template react
npm install tailwindcss @tailwindcss/vite framer-motion lucide-react
```

Step 2: Designing Theme Tokens (`src/index.css`)

Define dark slate themes and monospace fonts in Tailwind CSS v4:

```
@import "tailwindcss";

@theme {
  --color-bg-primary: #0a0e17;
  --color-bg-card: #111827;
  --color-bg-card-hover: #1f2937;
  --color-text-primary: #f3f4f6;
  --color-text-secondary: #9ca3af;
  --color-accent: #3b82f6;
  --color-border: #1f2937;
  --font-sans: 'Inter', sans-serif;
  --font-mono: 'JetBrains Mono', monospace;
}
```

Step 3: Centralizing Content (`src/constants/data.js`)

Store all personal bio details, project metadata, skills, and links in a single file:

```
export const PERSONAL = {
  name: 'Sk Sahil',
  role: 'Tech & Cybersecurity Enthusiast',
```

```
college: 'B.Tech Computer Science',
email: 'sksahil01018@gmail.com',
github: 'https://github.com/Sahil-9879',
linkedin: 'https://www.linkedin.com/in/sk-sahil-061a5a373/',
leetcode: 'https://leetcode.com/u/sahil_0205/',
resume: '/resume.pdf',
};
```

Step 4: Building Custom Hooks (`src/hooks/useTerminal.js`)

Implement command history, input event keydown listeners, and actions:

```
if (trimmed === 'resume') {
  const a = document.createElement('a');
  a.href = PERSONAL.resume;
  a.download = 'Sk_Sahil_Resume.pdf';
  a.click();
}
```

Step 5: Git Version Control & Deployment

```
git init
git add -A
git commit -m "feat: Sk Sahil developer portfolio"
git remote add origin https://github.com/Sahil-9879/sk_sahil_portfolio.git
git branch -M main
git push -u origin main
```